

RED HAWK TOOL – PENETRATION TESTING MANUAL

INTRODUCTION:

Red Hawk is an open-source web application reconnaissance and vulnerability scanning tool. It is primarily used in cybersecurity to gather information about target websites and identify potential security weaknesses. The tool performs various tasks such as DNS lookup, subdomain discovery, server information gathering, and vulnerability scanning. It is widely used during the initial phase of penetration testing to understand the structure and exposure of a target system.

REQUIREMENTS:

Before starting, ensure the following:

- Kali Linux installed or any Linux-based system
- Active Internet connection
- Git installed
- PHP installed

WORKING PRINCIPLE:

Red Hawk operates as an information gathering and scanning tool. It interacts with various external services, APIs, and system utilities to collect data about a target website.

1. The user inputs a target website.
2. The tool sends requests to gather DNS, IP, and server data.
3. It queries external APIs (e.g., Moz) for additional information.
4. It performs scans such as port scanning and vulnerability checks.
5. The collected data is displayed to the user for analysis.

INSTALLATION STEPS:

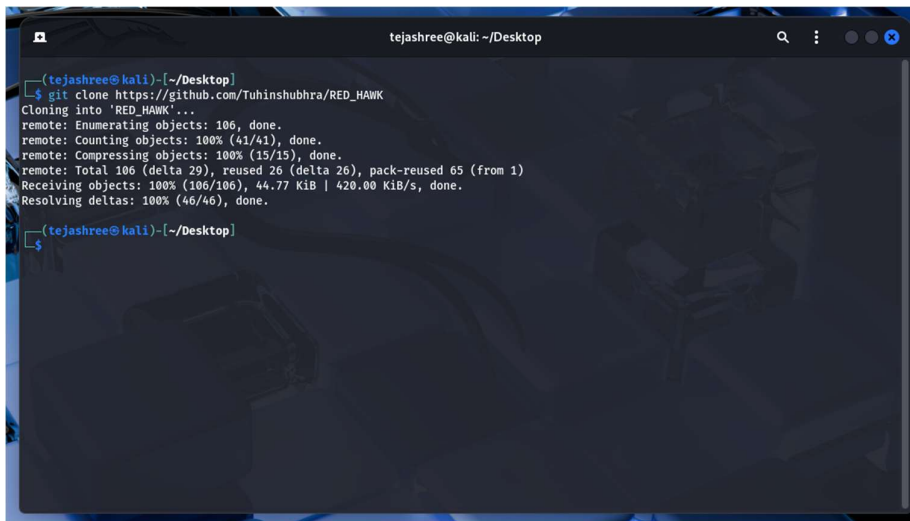
Step 1: Open Terminal

Launch Kali Linux and open the terminal.

Step 2: Clone the Repository

Type the following command in the terminal:

git clone https://github.com/Tuhinshubhra/RED_HAWK

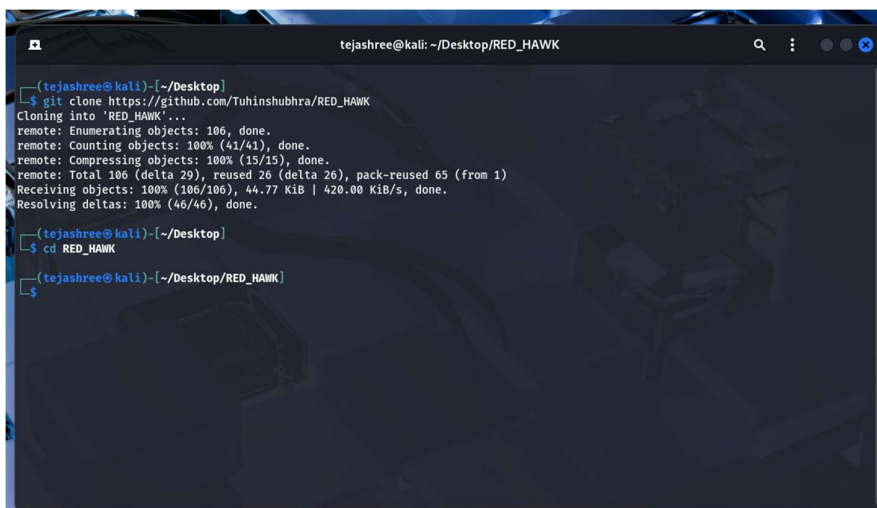
A terminal window titled 'tejashree@kali: ~/Desktop' showing the execution of the 'git clone' command. The output displays the progress of cloning the repository, including object enumeration, counting, and compression, all completed successfully.

```
(tejashree@kali)~[/Desktop]
$ git clone https://github.com/Tuhinshubhra/RED_HAWK
Cloning into 'RED_HAWK'...
remote: Enumerating objects: 106, done.
remote: Counting objects: 100% (41/41), done.
remote: Compressing objects: 100% (15/15), done.
remote: Total 106 (delta 20), reused 26 (delta 26), pack-reused 65 (from 1)
Receiving objects: 100% (106/106), 44.77 KiB | 420.00 KiB/s, done.
Resolving deltas: 100% (46/46), done.
(tejashree@kali)~[/Desktop]
$
```

Step 3: Go to the Directory

Type the following in the terminal:

cd RED_HAWK

A terminal window titled 'tejashree@kali: ~/Desktop/RED_HAWK' showing the execution of the 'cd' command. The prompt changes from '~[/Desktop]' to '~/Desktop/RED_HAWK', indicating the current directory has been updated.

```
(tejashree@kali)~[/Desktop]
$ git clone https://github.com/Tuhinshubhra/RED_HAWK
Cloning into 'RED_HAWK'...
remote: Enumerating objects: 106, done.
remote: Counting objects: 100% (41/41), done.
remote: Compressing objects: 100% (15/15), done.
remote: Total 106 (delta 20), reused 26 (delta 26), pack-reused 65 (from 1)
Receiving objects: 100% (106/106), 44.77 KiB | 420.00 KiB/s, done.
Resolving deltas: 100% (46/46), done.
(tejashree@kali)~[/Desktop]
$ cd RED_HAWK
(tejashree@kali)~[/Desktop/RED_HAWK]
$
```

Step 4: View Files in the Directory

List all files inside the directory to verify successful download:

ls

```
tejashree@kali: ~/Desktop/RED_HAWK

(tejashree@kali)-[~/Desktop]
└─$ git clone https://github.com/Tuhinshubhra/RED_HAWK
Cloning into 'RED_HAWK'...
remote: Enumerating objects: 106, done.
remote: Counting objects: 100% (41/41), done.
remote: Compressing objects: 100% (15/15), done.
remote: Total 106 (delta 29), reused 26 (delta 26), pack-reused 65 (from 1)
Receiving objects: 100% (106/106), 44.77 KiB | 420.00 KiB/s, done.
Resolving deltas: 100% (46/46), done.

(tejashree@kali)-[~/Desktop]
└─$ cd RED_HAWK

(tejashree@kali)-[~/Desktop/RED_HAWK]
└─$ ls
config.php  crawl  Dockerfile  functions.php  LICENSE  README.md  rhawk.php  sqlerrors.ini  var.php  version.txt

(tejashree@kali)-[~/Desktop/RED_HAWK]
└─$
```

Step 5: Run the Tool

Execute the tool using PHP:

php rhawk.php

```
tejashree@kali: ~/Desktop/RED_HAWK

(tejashree@kali)-[~/Desktop/RED_HAWK]
└─$ php rhawk.php

All In One Tool For Information Gathering And Vulnerability Scanning

RED_HAWK Ver 2.0.0
(C) Coded By - R3D0R@K0R_2H1N A.K.A Tuhinshubhra
[!] Shout Out - You ;)

[!] Enter The Website You Want To Scan : 
```

USING THE RED HAWK TOOL

Step 1: Enter Target Website

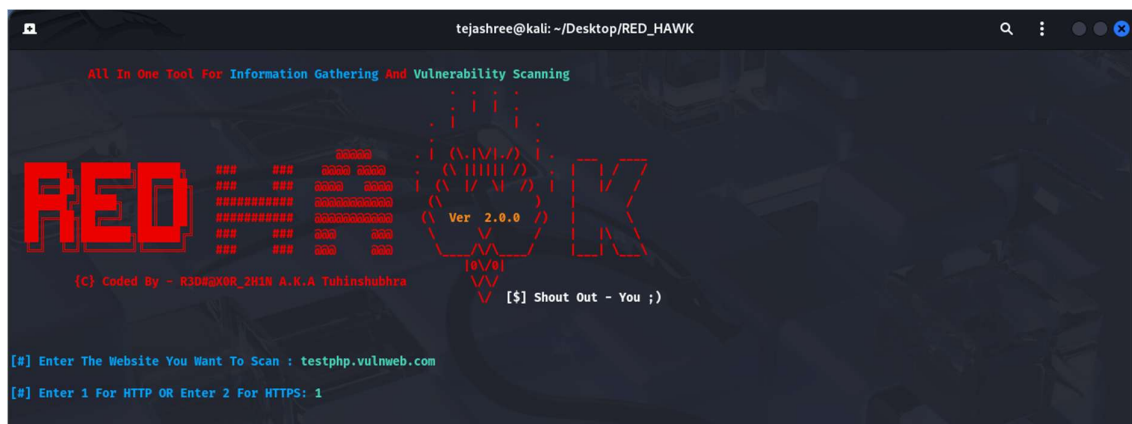
After launching the tool, the user will be prompted to enter a target website URL



Step 2: Select Protocol

Choose the protocol to be used:

- 1 - HTTP
- 2 - HTTPS



Step 3: Select Scan Type

The tool displays a list of available scanning options.

```
tejashree@kali: ~/Desktop/RED_HAWK

+-----+
+-----+ List Of Scans Or Actions +-----+
+-----+

Scanning Site : http://testphp.vulnweb.com

[0] Basic Recon (Site Title, IP Address, CMS, Cloudflare Detection, Robots.txt Scanner)
[1] Whois Lookup
[2] Geo-IP Lookup
[3] Grab Banners
[4] DNS Lookup
[5] Subnet Calculator
[6] NMAP Port Scan
[7] Subdomain Scanner
[8] Reverse IP Lookup & CMS Detection
[9] SQLi Scanner (Finds Links With Parameter And Scans For Error Based SQLi)
[10] Bloggers View (Information That Bloggers Might Be Interested In)
[11] WordPress Scan (Only If The Target Site Runs On WP)
[12] Crawler
[13] MX Lookup
[A] Scan For Everything - (The Old Lane Scanner)
[F] Fix (Checks For Required Modules and Installs Missing Ones)
[U] Check For Updates
[B] Scan Another Website (Back To Site Selection)
[Q] Quit!

[#] Choose Any Scan OR Action From The Above List: █
```

Step 4: Choose a Scan Option

Select the desired scan type

1. Basic Recon

DESCRIPTION:

Basic Recon (Basic Reconnaissance) is used to collect general information about a target website. It gathers details such as IP address, server information, CMS detection (Content Management System), and security-related data.

```
tejashree@kali: ~/Desktop/RED_HAWK

[0] Basic Recon (Site Title, IP Address, CMS, Cloudflare Detection, Robots.txt Scanner)
[1] Whois Lookup
[2] Geo-IP Lookup
[3] Grab Banners
[4] DNS Lookup
[5] Subnet Calculator
[6] NMAP Port Scan
[7] Subdomain Scanner
[8] Reverse IP Lookup & CMS Detection
[9] SQLi Scanner (Finds Links With Parameter And Scans For Error Based SQLi)
[10] Bloggers View (Information That Bloggers Might Be Interested In)
[11] WordPress Scan (Only If The Target Site Runs On WP)
[12] Crawler
[13] MX Lookup
[A] Scan For Everything - (The Old Lane Scanner)
[F] Fix (Checks For Required Modules and Installs Missing Ones)
[U] Check For Updates
[B] Scan Another Website (Back To Site Selection)
[Q] Quit!

[#] Choose Any Scan OR Action From The Above List: 0

[*] Scanning Begins ...
[1] Scanning Site: http://testphp.vulnweb.com
[S] Scan Type : BASIC SCAN

[INFO] Site Title:
[INFO] IP address: 44.228.249.3
[INFO] Web Servers: Could Not Detect
[INFO] CMS: Could Not Detect
[INFO] Cloudflare: Not Detected
[INFO] Robots File: Could NOT Find robots.txt!

[*] Scanning Complete. Press Enter To Continue OR CTRL + C To Stop
```

RESULT:

The tool displays available information about the target system. Some parameters may not be detected depending on the website configuration.

USE CASE:

Initial information gathering during the reconnaissance phase of penetration testing.

LIMITATION:

The accuracy of results depends on the target server's configuration and security settings.

2. Whois Lookup

DESCRIPTION

Whois Lookup is used to retrieve domain registration details such as the domain owner, registrar, registration date, and expiry date.



```
tejashree@kali: ~/Desktop/RED_HAWK
+-----+-----+
| List Of Scans Or Actions |
+-----+-----+
Scanning Site : http://testphp.vulnweb.com

[0] Basic Recon (Site Title, IP Address, CMS, Cloudflare Detection, Robots.txt Scanner)
[1] Whois Lookup
[2] Geo-IP Lookup
[3] Grab Banners
[4] DNS Lookup
[5] Subnet Calculator
[6] NMAP Port Scan
[7] Subdomain Scanner
[8] Reverse IP Lookup & CMS Detection
[9] SQLi Scanner (Finds Links With Parameter And Scans For Error Based SQLi)
[10] Bloggers View (Information That Bloggers Might Be Interested In)
[11] WordPress Scan (Only If The Target Site Runs On WP)
[12] Crawler
[13] MX Lookup
[A] Scan For Everything - (The Old Lane Scanner)
[F] Fix (Checks For Required Modules and Installs Missing Ones)
[U] Check For Updates
[B] Scan Another Website (Back To Site Selection)
[Q] Quit!

[#] Choose Any Scan OR Action From The Above List: 1

[+] Scanning Begins ...
[i] Scanning Site: http://testphp.vulnweb.com
[S] Scan Type : WHOIS Lookup
[~] Whois Lookup Result:
error valid key required

[*] Scanning Complete. Press Enter To Continue OR CTRL + C To Stop
```

RESULT:

The tool was unable to retrieve the information due to missing API credentials.

USE CASE:

Used to identify domain ownership and registration details during reconnaissance.

LIMITATION:

This feature requires a paid API service. Without valid credentials, it will not function.

3. Geo-IP Lookup

DESCRIPTION:

Geo-IP Lookup is used to identify the geographical location of a target website based on its IP address. It provides details such as country, region, city, and Internet Service Provider (ISP).

```
tejashtree@kali: ~/Desktop/RED_HAWK
+-----+
+ List Of Scans Or Actions +
+-----+

Scanning Site : http://testphp.vulnweb.com

[0] Basic Recon (Site Title, IP Address, CMS, Cloudflare Detection, Robots.txt Scanner)
[1] Whois Lookup
[2] Geo-IP Lookup
[3] Grab Banners
[4] DNS Lookup
[5] Subnet Calculator
[6] NMAP Port Scan
[7] Subdomain Scanner
[8] Reverse IP Lookup & CMS Detection
[9] SQLi Scanner (Finds Links With Parameter And Scans For Error Based SQLi)
[10] Bloggers View (Information That Bloggers Might Be Interested In)
[11] WordPress Scan (Only If The Target Site Runs On WP)
[12] Crawler
[13] MX Lookup
[A] Scan For Everything - (The Old Lame Scanner)
[F] Fix (Checks For Required Modules and Installs Missing Ones)
[U] Check For Updates
[B] Scan Another Website (Back To Site Selection)
[Q] Quit!

[*] Choose Any Scan OR Action From The Above List: 2

[*] Scanning Begins ...
[1] Scanning Site: http://testphp.vulnweb.com
[5] Scan Type : GEO-IP Lookup

[GEO-IP] IP: 44.228.249.3
[GEO-IP] Country: United States
[GEO-IP] State: Oregon
[GEO-IP] City: Boardman
[GEO-IP] Latitude: 45.8401
[GEO-IP] Longitude: -119.705

[*] Scanning Complete. Press Enter To Continue OR CTRL + C To Stop
```

RESULT:

The tool successfully retrieved the geographical information of the target website, including its location and ISP details.

USE CASE:

Used to determine the physical location and hosting details of a target system during reconnaissance.

LIMITATION:

The accuracy of the location depends on the IP database and may not always be precise.

4. Banner Grabbing

DESCRIPTION:

Banner Grabbing is used to extract information about the web server and services running on the target system. It helps identify the server type and underlying technologies.



```
tejashree@kali: ~/Desktop/RED_HAWK
+-----+
+ List Of Scans Or Actions                               +
+-----+
Scanning Site : http://testphp.vulnweb.com

[0] Basic Recon (Site Title, IP Address, CMS, Cloudflare Detection, Robots.txt Scanner)
[1] Whois Lookup
[2] Geo-IP Lookup
[3] Grab Banners
[4] DNS Lookup
[5] Subnet Calculator
[6] NMAP Port Scan
[7] Subdomain Scanner
[8] Reverse IP Lookup & CMS Detection
[9] SQLi Scanner (Finds Links With Parameter And Scans For Error Based SQLi)
[10] Bloggers View (Information That Bloggers Might Be Interested In)
[11] WordPress Scan (Only If The Target Site Runs On WP)
[12] Crawler
[13] MX Lookup
[A] Scan For Everything - (The Old Lame Scanner)
[F] Fix (Checks For Required Modules and Installs Missing Ones)
[U] Check For Updates
[B] Scan Another Website (Back To Site Selection)
[Q] Quit

[0] Choose Any Scan OR Action From The Above List: 3

[*] Scanning Begins ...
[*] Scanning Site: http://testphp.vulnweb.com
[*] Scan Type : Banner Grabbing

[*] Scanning Complete. Press Enter To Continue OR CTRL + C To Stop
```

RESULT:

The tool attempted to retrieve server information, but no details were disclosed by the target. This indicates that the server is configured to hide sensitive information.

USE CASE:

Used to identify server technologies and services for further security analysis.

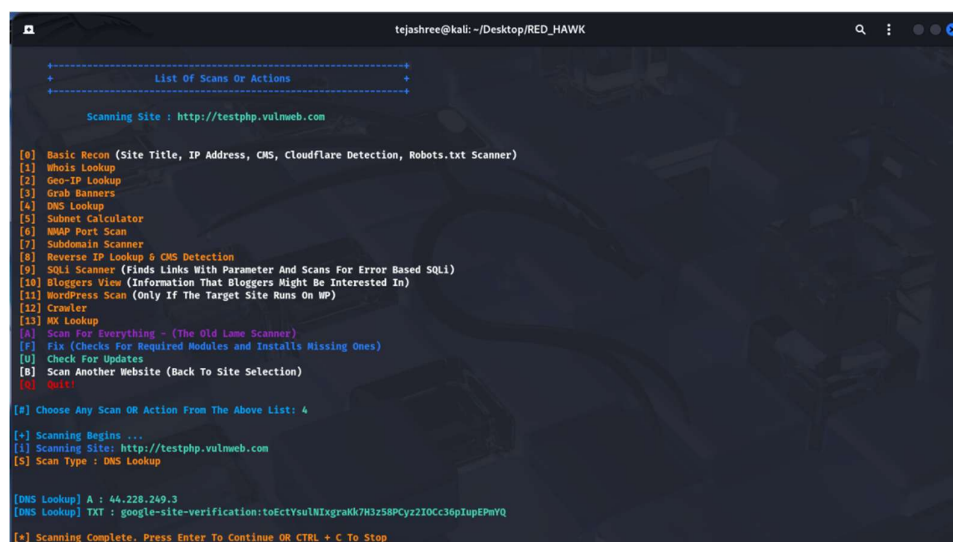
LIMITATION:

Many modern servers disable or hide banner information for security reasons, so results may not always be available.

5. DNS Lookup

DESCRIPTION:

DNS Lookup is used to retrieve domain-related records such as IP address mappings, name servers, and mail exchange (MX) records associated with a target website.



```
tejashree@kali: ~/Desktop/RED_HAWK
+-----+
+-----+ List Of Scans Or Actions +-----+
+-----+

Scanning Site : http://testphp.vulnweb.com

[0] Basic Recon (Site Title, IP Address, CMS, Cloudflare Detection, Robots.txt Scanner)
[1] Whois Lookup
[2] Geo-IP Lookup
[3] Grab Banners
[4] DNS Lookup
[5] Subnet Calculator
[6] NMAP Port Scan
[7] Subdomain Scanner
[8] Reverse IP Lookup & CMS Detection
[9] SQLi Scanner (Finds Links With Parameter And Scans For Error Based SQLi)
[10] Bloggers View (Information That Bloggers Might Be Interested In)
[11] WordPress Scan (Only If The Target Site Runs On WP)
[12] Crawler
[13] MX Lookup
[a] Scan For Everything - (The Old Lane Scanner)
[F] Fix (Checks For Required Modules and Installs Missing Ones)
[U] Check For Updates
[B] Scan Another Website (Back To Site Selection)
[q] quit

[*] Choose Any Scan OR Action From The Above List: 4

[*] Scanning Begins ...
[1] Scanning Site: http://testphp.vulnweb.com
[S] Scan Type : DNS Lookup

[DNS Lookup] A : 44.228.249.3
[DNS Lookup] TXT : google-site-verification:toEctYsu1NIXgraK37H3z58PCy22IOcc36pTupEPwYQ

[*] Scanning Complete. Press Enter To Continue OR CTRL + C To Stop
```

RESULT:

The tool successfully retrieved DNS records for the target website. The output included details such as IP address and other domain configuration records.

USE CASE:

Used to understand the domain configuration and identify associated services during reconnaissance.

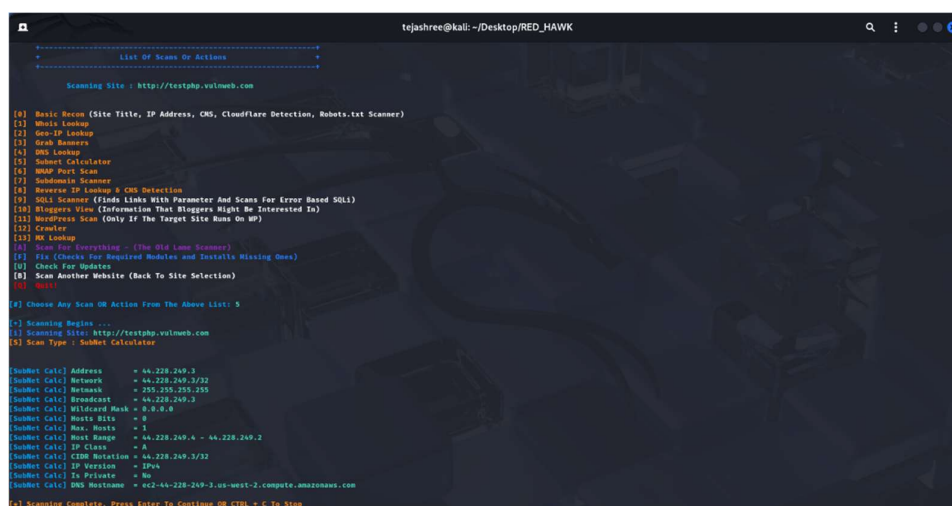
LIMITATION:

Some DNS records may be restricted or hidden depending on the server configuration.

6. Subnet Calculator

DESCRIPTION:

The Subnet Calculator is used to divide a network into smaller sub-networks (subnets) and calculate details such as IP range, subnet mask, and number of hosts.



```
tejasree@kali: ~/Desktop/RED_HAWK
+-----+
+ List Of Scans Or Actions +
+-----+

Scanning Site : http://testphp.vulnweb.com

[0] Basic Recon (Site Title, IP Address, CMS, Cloudflare Detection, Robots.txt Scanner)
[1] Basic Lookup
[2] Geo-IP Lookup
[3] Grab Banners
[4] DNS Lookup
[5] Subnet Calculator
[6] WMAP Port Scan
[7] Subdomain Scanner
[8] Reverse IP Lookup & CMS Detection
[9] SQLi Scanner (Finds Links With Parameter And Scans For Error Based SQLi)
[10] Bloggers View (Information That Bloggers Might Be Interested In)
[11] WordPress Scan (Only If The Target Site Runs On WP)
[12] Crawler
[13] NX Lookup
[14] Scan For Everything - (The Old Line Scanner)
[15] Fix (Checks For Required Modules And Installs Missing Ones)
[16] Check For Updates
[17] Scan Another Website (Back To Site Selection)
[18] Exit

[?] Choose Any Scan Or Action From The Above List: 5

[?] Scanning Begins ...
[1] Scanning Site: http://testphp.vulnweb.com
[5] Scan Type : Subnet Calculator

Subnet Calc| Address      = 44.228.249.3
Subnet Calc| Network     = 44.228.249.3/22
Subnet Calc| Netmask      = 255.255.255.255
Subnet Calc| Broadcast    = 44.228.249.3
Subnet Calc| Wildcard Mask = 0.0.0.0
Subnet Calc| Hosts Bits    = 8
Subnet Calc| Hosts        = 2
Subnet Calc| Host Range   = 44.228.249.4 - 44.228.249.2
Subnet Calc| IP Class     = A
Subnet Calc| CIDR Notation = 44.228.249.3/22
Subnet Calc| IP Version   = IPv4
Subnet Calc| Is Private    = No
Subnet Calc| DNS Hostname  = ec2-44-228-249-3.us-west-2.compute.amazonaws.com

[?] Scanning Complete. Press Enter To Continue OR CTRL + C To Stop
```

RESULT:

The tool calculated subnet details for the given input, including IP range, subnet mask, and total number of hosts.

USE CASE:

Used to analyze network structure and understand IP allocation during reconnaissance.

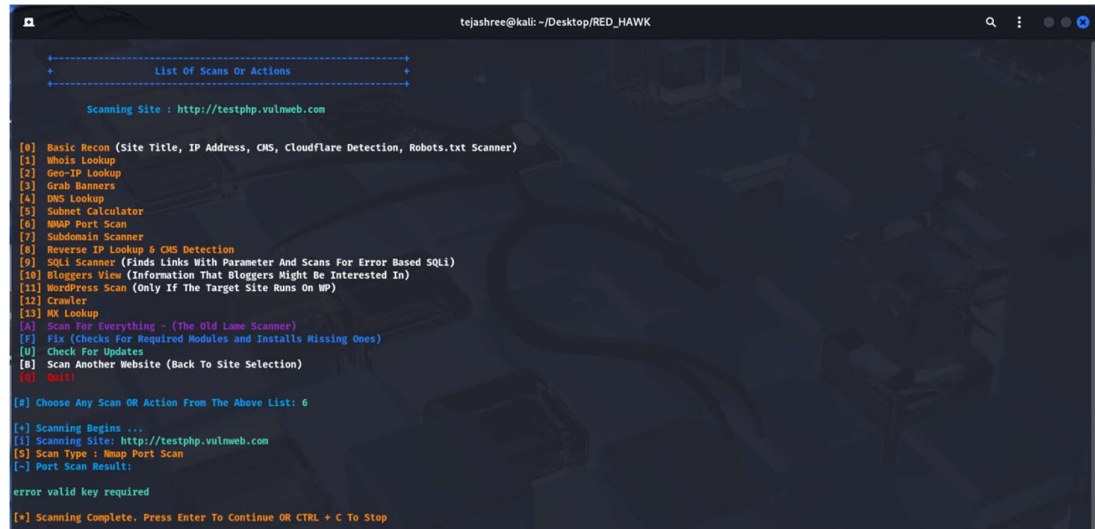
LIMITATION:

The accuracy of results depends on correct input values, and this feature is mainly useful for network-level analysis rather than web application scanning.

7. NMAP Port Scan

DESCRIPTION:

Nmap Port Scan is used to identify open ports and services running on a target system. It helps in detecting exposed services that may be vulnerable to attacks.



```
tejasree@kali: ~/Desktop/RED_HAWK
+-----+
+-----+ List Of Scans Or Actions +-----+
+-----+
Scanning Site : http://testphp.vulnweb.com

[0] Basic Recon (Site Title, IP Address, CMS, Cloudflare Detection, Robots.txt Scanner)
[1] Whois Lookup
[2] Geo-IP Lookup
[3] Grab Banners
[4] DNS Lookup
[5] Subnet Calculator
[6] NMAP Port Scan
[7] Subdomain Scanner
[8] Reverse IP Lookup & CMS Detection
[9] SQL Scanner (Finds Links With Parameter And Scans For Error Based SQLi)
[10] Bloggers View (Information That Bloggers Might Be Interested In)
[11] WordPress Scan (Only If The Target Site Runs On WP)
[12] Crawler
[13] XM Lookup
[14] Scan For Everything - (The Old Lane Scanner)
[F] Fix (Checks For Required Modules and Installs Missing Ones)
[U] Check For Updates
[R] Scan Another Website (Back To Site Selection)
[Q] Quit

[a] Choose Any Scan OR Action From The Above List: 6

[*] Scanning Begins ...
[1] Scanning Site: http://testphp.vulnweb.com
[5] Scan Type : Nmap Port Scan
[-] Port Scan Result:
error valid key required
[*] Scanning Complete. Press Enter To Continue OR CTRL + C To Stop
```

RESULT:

The tool attempted to perform a port scan, but the scan could not be completed due to missing API configuration or required permissions.

USE CASE:

Used to identify open ports and services during the reconnaissance and scanning phase of penetration testing.

LIMITATION:

This feature may require proper configuration or API access. Without it, the scan may not execute successfully.

8. Subdomain Scanner

DESCRIPTION:

The Subdomain Scanner is used to discover subdomains associated with a target domain. Subdomains help identify different sections or services of a website.

A screenshot of a terminal window titled 'tejashree@kali: ~/Desktop/RED_HAWK'. The terminal shows the 'List Of Scans Or Actions' menu with various options. Option 7, 'Subdomain Scanner', is selected. The scanning site is set to 'http://testphp.vulnweb.com'. The scan type is 'Subdomain Scanner'. The result shows 'Total Subdomains Found : 0'. The terminal text is as follows:

```
tejashree@kali: ~/Desktop/RED_HAWK
+-----+
+-----+ List Of Scans Or Actions +-----+
+-----+
Scanning Site : http://testphp.vulnweb.com

[0] Basic Recon (Site Title, IP Address, CMS, Cloudflare Detection, Robots.txt Scanner)
[1] Whois Lookup
[2] Geo-IP Lookup
[3] Grab Banners
[4] DNS Lookup
[5] Subnet Calculator
[6] NMAP Port Scan
[7] Subdomain Scanner
[8] Reverse IP Lookup & CMS Detection
[9] SQLi Scanner (Finds Links With Parameter And Scans For Error Based SQLi)
[10] Bloggers View (Information That Bloggers Might Be Interested In)
[11] WordPress Scan (Only If The Target Site Runs On WP)
[12] Crawler
[13] MX Lookup
[A] Scan For Everything - (The Old Lame Scanner)
[F] Fix (Checks For Required Modules and Installs Missing Ones)
[U] Check For Updates
[B] Scan Another Website (Back To Site Selection)
[0] Null!

[a] Choose Any Scan OR Action From The Above List: 7

[*] Scanning Begins ...
[i] Scanning Site: http://testphp.vulnweb.com
[S] Scan Type : Subdomain Scanner
[i] Total Subdomains Found : 0

[*] Scanning Complete. Press Enter To Continue OR CTRL + C To Stop
```

RESULT:

The tool executed the scan but did not discover any subdomains for the target website. This may occur if subdomains are not publicly available or are hidden.

USE CASE:

Used to expand the attack surface by identifying additional subdomains during reconnaissance.

LIMITATION:

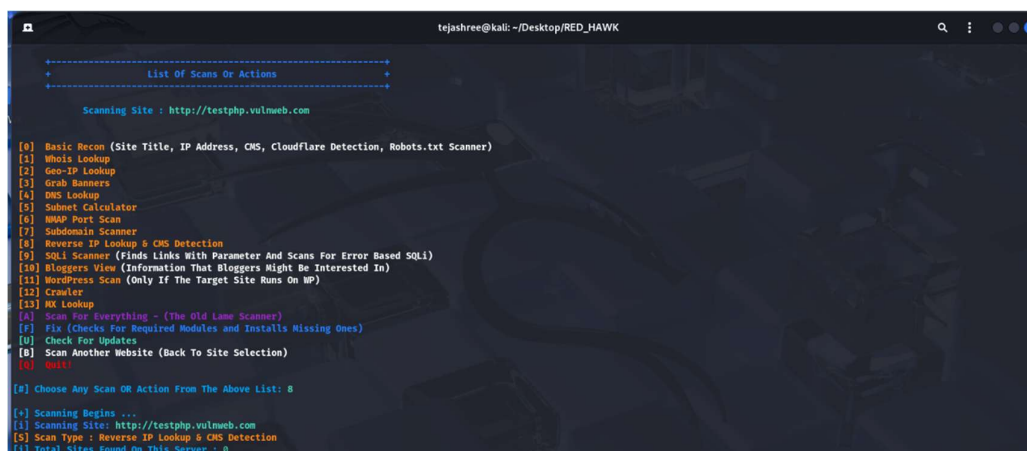
The tool may not detect all subdomains, especially if they are hidden, restricted, or not indexed.

9. Reverse IP Lookup & CMS Detection

DESCRIPTION:

Reverse IP Lookup is used to identify other domains hosted on the same server (IP address) as the target website.

CMS Detection is used to identify the Content Management System (CMS) used by the website, such as WordPress, Joomla or Drupal.



```
tejashee@kali: ~/Desktop/RED_HAWK
+-----+
+-----+ List Of Scans Or Actions +-----+
+-----+
Scanning Site : http://testphp.vulnweb.com

[0] Basic Recon (Site Title, IP Address, CMS, CloudFlare Detection, Robots.txt Scanner)
[1] Whois Lookup
[2] Geo-IP Lookup
[3] Grab Banners
[4] DNS Lookup
[5] Subnet Calculator
[6] NMAP Port Scan
[7] Subdomain Scanner
[8] Reverse IP Lookup & CMS Detection
[9] SQL Scanner (Finds Links With Parameter And Scans For Error Based SQLi)
[10] Bloggers View (Information That Bloggers Might Be Interested In)
[11] WordPress Scan (Only If The Target Site Runs On WP)
[12] Crawler
[13] RX Lookup
[14] Scan For Everything - (The Old Lane Scanner)
[F] Fix (Checks For Required Modules and Installs Missing Ones)
[U] Check For Updates
[B] Scan Another Website (Back To Site Selection)
[Q] Quit

[8] Choose Any Scan OR Action From The Above List: 8

[*] Scanning Begins ...
[1] Scanning Site: http://testphp.vulnweb.com
[5] Scan Type : Reverse IP Lookup & CMS Detection
[1] Total Sites Found On This Server : 0
```

RESULT:

The tool attempted to perform Reverse IP Lookup and CMS Detection. However, limited or no results were obtained. This may be due to restricted data access or the website hiding its technology details.

USE CASE:

Reverse IP Lookup helps identify other domains sharing the same server, which can reveal related targets.

CMS Detection helps identify the platform used, which is useful for identifying CMS-specific vulnerabilities.

LIMITATION:

Reverse IP Lookup may return no results if no other domains are hosted on the same IP or if data is restricted.

CMS Detection may fail if the website hides its technology stack or uses security protections.

10. SQL Vulnerability Scanner

DESCRIPTION:

The SQL Vulnerability Scanner is used to check for potential SQL Injection vulnerabilities in a target website. SQL Injection is an attack technique where malicious queries are inserted into input fields to manipulate the database.

```
[0] echo %date% : %date%>%cd%\%date%.txt
[1] echo %date% : %date%>\\192.168.0.1\temp\com
[2] echo %date% ...
[3]
[4]
[5]
[6]
[7]
[8]
[9]
[10]
[11]
[12]
[13]
[14]
[15]
[16]
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```

RESULT:

The tool performed the scan but did not return any confirmed vulnerabilities. This may be due to secure configurations or limitations of the tool.

USE CASE:

Used to identify possible SQL Injection points in a web application during security testing.

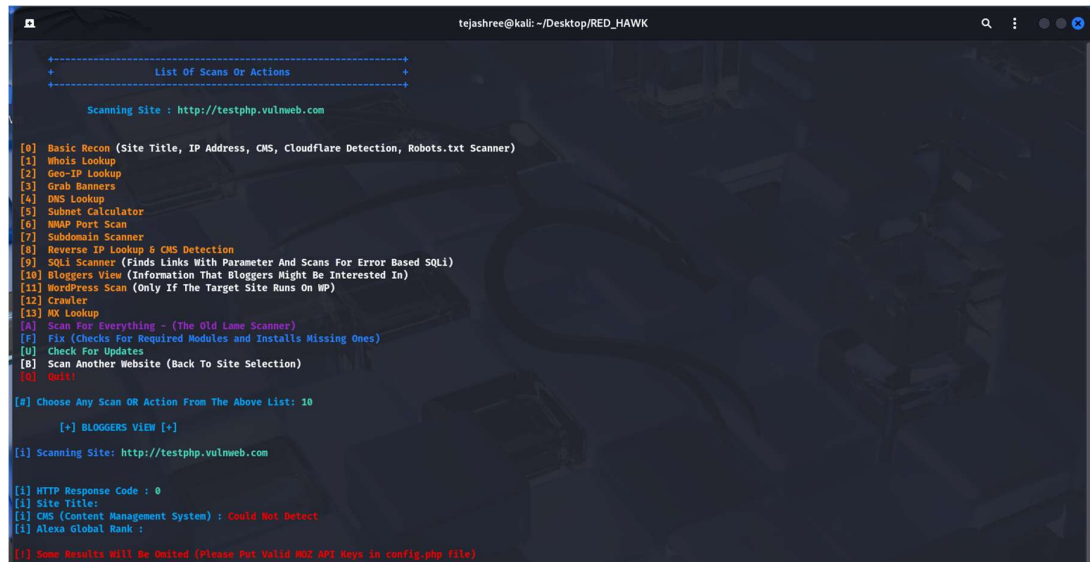
LIMITATION:

The tool may not detect all vulnerabilities and cannot guarantee complete security. Accurate detection often requires manual testing and advanced tools.

11. Bloggers View

DESCRIPTION:

Bloggers View is used to retrieve SEO-related information about a website, such as domain authority, page authority, backlinks, and overall popularity metrics.



```
tejashree@kali: ~/Desktop/RED_HAWK

+-----+
+ List Of Scans Or Actions                               +
+-----+

Scanning Site : http://testphp.vulnweb.com

[0] Basic Recon (Site Title, IP Address, CMS, Cloudflare Detection, Robots.txt Scanner)
[1] Whois Lookup
[2] Geo-IP Lookup
[3] Grab Banners
[4] DNS Lookup
[5] Subnet Calculator
[6] NMAP Port Scan
[7] Subdomain Scanner
[8] Reverse IP Lookup & CMS Detection
[9] SQLi Scanner (Finds Links With Parameter And Scans For Error Based SQLi)
[10] Bloggers View (Information That Bloggers Might Be Interested In)
[11] WordPress Scan (Only If The Target Site Runs On WP)
[12] Crawler
[13] RX Lookup
[A] Scan For Everything - (The Old Lane Scanner)
[F] Fix (Checks For Required Modules and Installs Missing Ones)
[U] Check For Updates
[B] Scan Another Website (Back To Site Selection)
[Q] Quit

[8] Choose Any Scan OR Action From The Above List: 10

[+] BLOGGERS VIEW [+]

[1] Scanning Site: http://testphp.vulnweb.com

[1] HTTP Response Code : 0
[1] Site Title:
[1] CMS (Content Management System) : Could Not Detect
[1] Alexa Global Rank :

[!] Some Results Will Be Omitted (Please Put Valid MOZ API Keys in config.php file)
```

RESULT:

The tool attempted to retrieve SEO information but was unable to produce results due to missing API credentials.

USE CASE:

Used to analyze the popularity and search engine performance of a website during reconnaissance.

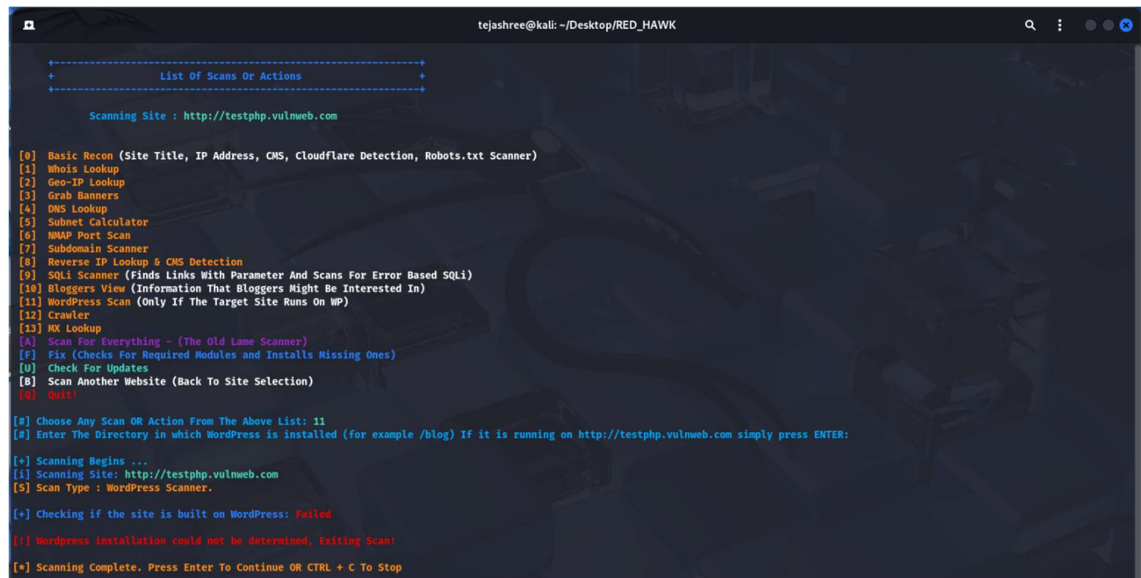
LIMITATION:

This feature requires a paid API service (e.g., Moz API). Without valid credentials, it will not function.

12. WordPress Scan

DESCRIPTION:

WordPress Scan is used to check whether a website is built using WordPress and to identify its components such as themes, plugins, and potential vulnerabilities.



```
tejashree@kali: ~/Desktop/RED_HAWK

+-----+
+-----+ List Of Scans Or Actions +-----+
+-----+

Scanning Site : http://testphp.vulnweb.com

[0] Basic Recon (Site Title, IP Address, CMS, Cloudflare Detection, Robots.txt Scanner)
[1] Whois Lookup
[2] Geo-IP Lookup
[3] Grab Banners
[4] DNS Lookup
[5] Subnet Calculator
[6] NMAP Port Scan
[7] Subdomain Scanner
[8] Reverse IP Lookup & CMS Detection
[9] SQLi Scanner (Finds Links With Parameter And Scans For Error Based SQLi)
[10] Bloggers View (Information That Bloggers Might Be Interested In)
[11] WordPress Scan (Only If The Target Site Runs On WP)
[12] Crawler
[13] MX Lookup
[A] Scan For Everything - (The Old Lame Scanner)
[F] Fix (Checks For Required Modules and Installs Missing Ones)
[U] Check For Updates
[B] Scan Another Website (Back To Site Selection)
[Q] Quit!

[*] Choose Any Scan OR Action From The Above List: 11
[*] Enter The Directory in which WordPress is installed (for example /blog) If it is running on http://testphp.vulnweb.com simply press ENTER:

[*] Scanning Begins ...
[*] Scanning Site: http://testphp.vulnweb.com
[*] Scan Type : WordPress Scanner.

[*] Checking if the site is built on WordPress: Failed
[*] Wordpress installation could not be determined, Exiting Scan!
[*] Scanning Complete. Press Enter To Continue OR CTRL + C To Stop
```

RESULT:

The tool attempted to detect WordPress components but returned limited or no information. This may be due to the website not using WordPress or hiding its technology details.

USE CASE:

Used to identify WordPress-based websites and analyze their components for potential security issues.

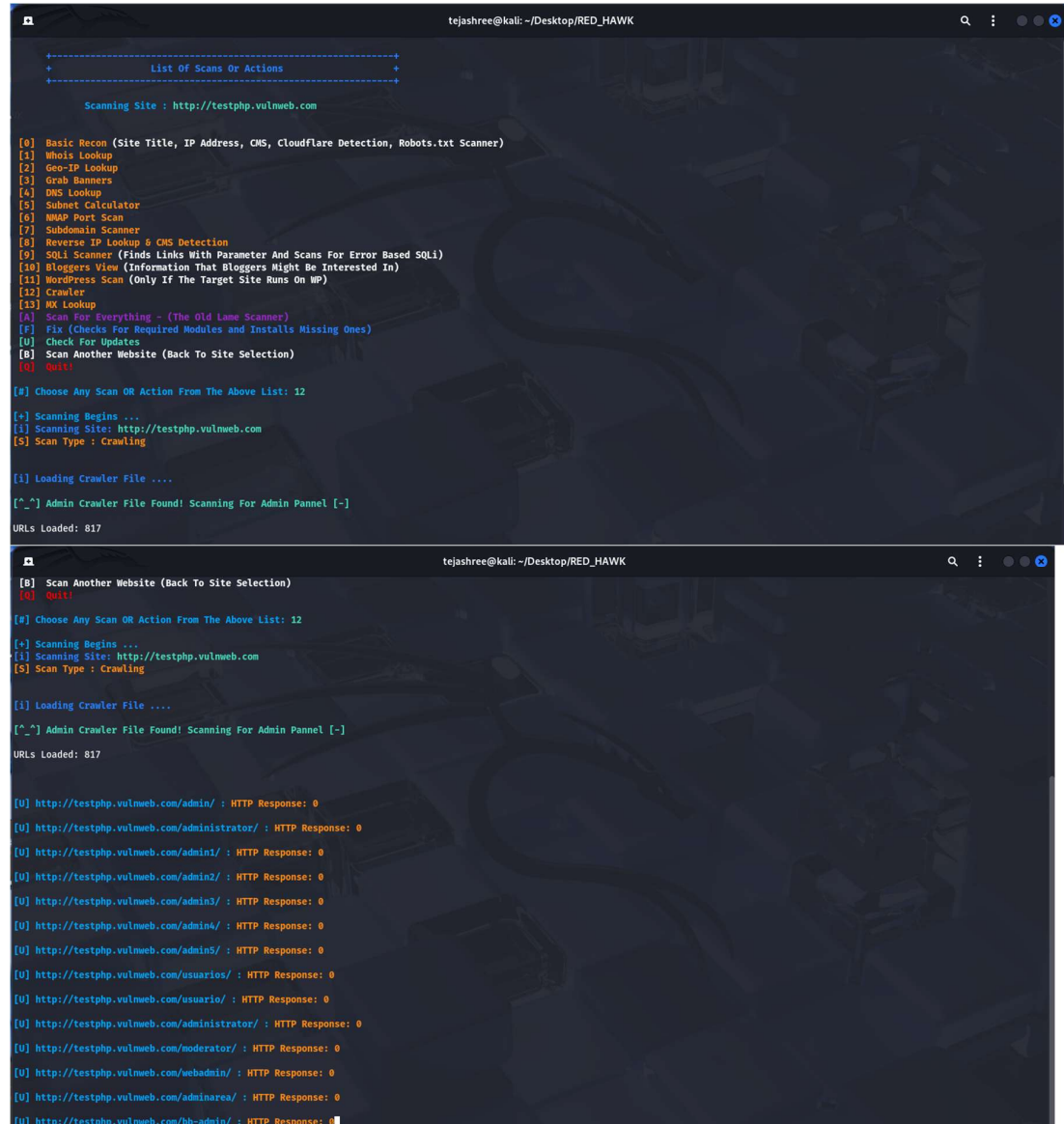
LIMITATION:

The scan may not detect WordPress if the website hides its structure or uses security measures to prevent detection.

13. Crawler

DESCRIPTION:

A Crawler (web crawler) is used to automatically browse a website and collect URLs by following links from one page to another.



```
tejashree@kali: ~/Desktop/RED_HAWK

+-----+
+-----+ List Of Scans Or Actions +-----+
+-----+

Scanning Site : http://testphp.vulnweb.com

[0] Basic Recon (Site Title, IP Address, CMS, Cloudflare Detection, Robots.txt Scanner)
[1] Whois Lookup
[2] Geo-IP Lookup
[3] Grab Banners
[4] DNS Lookup
[5] Subnet Calculator
[6] NMAP Port Scan
[7] Subdomain Scanner
[8] Reverse IP Lookup & CMS Detection
[9] SQLi Scanner (Finds Links With Parameter And Scans For Error Based SQLi)
[10] Bloggers View (Information That Bloggers Might Be Interested In)
[11] Wordpress Scan (Only If The Target Site Runs On WP)
[12] Crawler
[13] MX Lookup
[A] Scan For Everything - (The Old Lame Scanner)
[F] Fix (Checks For Required Modules and Installs Missing Ones)
[0] Check For Updates
[B] Scan Another Website (Back To Site Selection)
[Q] Quit

[8] Choose Any Scan OR Action From The Above List: 12

[+] Scanning Begins ...
[i] Scanning Site: http://testphp.vulnweb.com
[S] Scan Type : Crawling

[i] Loading Crawler File ....
[*_*] Admin Crawler File Found! Scanning For Admin Pannel [-]
URLs Loaded: 817

tejashree@kali: ~/Desktop/RED_HAWK

[B] Scan Another Website (Back To Site Selection)
[Q] Quit

[8] Choose Any Scan OR Action From The Above List: 12

[+] Scanning Begins ...
[i] Scanning Site: http://testphp.vulnweb.com
[S] Scan Type : Crawling

[i] Loading Crawler File ....
[*_*] Admin Crawler File Found! Scanning For Admin Pannel [-]
URLs Loaded: 817

[U] http://testphp.vulnweb.com/admin/ : HTTP Response: 0
[U] http://testphp.vulnweb.com/administrator/ : HTTP Response: 0
[U] http://testphp.vulnweb.com/admin1/ : HTTP Response: 0
[U] http://testphp.vulnweb.com/admin2/ : HTTP Response: 0
[U] http://testphp.vulnweb.com/admin3/ : HTTP Response: 0
[U] http://testphp.vulnweb.com/admin4/ : HTTP Response: 0
[U] http://testphp.vulnweb.com/admin5/ : HTTP Response: 0
[U] http://testphp.vulnweb.com/usuarios/ : HTTP Response: 0
[U] http://testphp.vulnweb.com/usuario/ : HTTP Response: 0
[U] http://testphp.vulnweb.com/administrator/ : HTTP Response: 0
[U] http://testphp.vulnweb.com/moderator/ : HTTP Response: 0
[U] http://testphp.vulnweb.com/webadmin/ : HTTP Response: 0
[U] http://testphp.vulnweb.com/adminarea/ : HTTP Response: 0
[U] http://testphp.vulnweb.com/bb-admin/ : HTTP Response: 0
```

RESULT:

The crawler was successfully executed and retrieved multiple URLs from the target website, demonstrating its ability to map the site structure.

USE CASE:

Used to discover hidden pages and understand the structure of a website during reconnaissance.

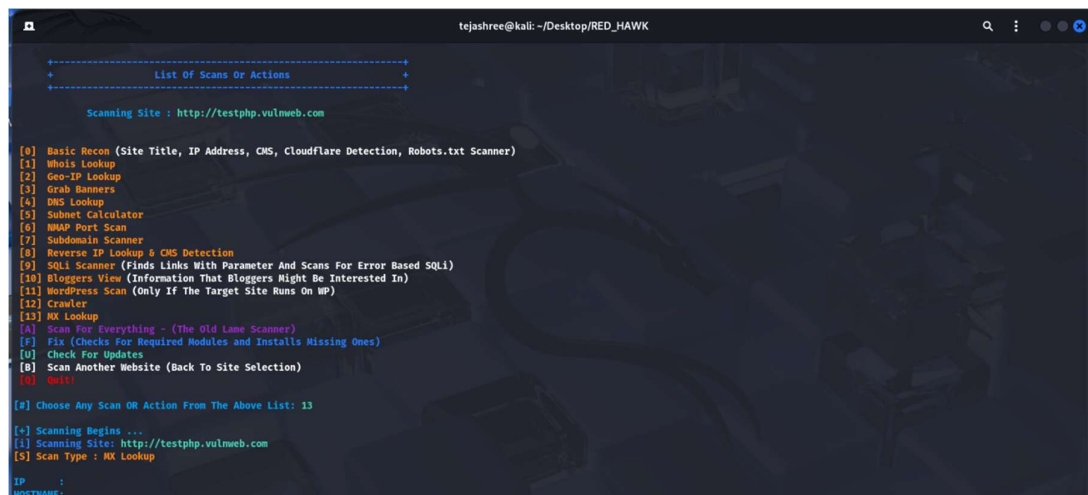
LIMITATION:

The crawler may not access restricted or protected pages and its performance depends on the website's structure and response time.

14. MX Lookup

DESCRIPTION:

MX (Mail Exchange) Lookup is used to identify the mail servers responsible for receiving emails for a domain.



```
tejashree@kali: ~/Desktop/RED_HAWK
+-----+
+ List Of Scans Or Actions                               +
+-----+
Scanning Site : http://testphp.vulnweb.com

[0] Basic Recon (Site Title, IP Address, CMS, Cloudflare Detection, Robots.txt Scanner)
[1] Whois Lookup
[2] Geo-IP Lookup
[3] Grab Banners
[4] DNS Lookup
[5] Subnet Calculator
[6] NMAP Port Scan
[7] Subdomain Scanner
[8] Reverse IP Lookup & CMS Detection
[9] SQLi Scanner (Finds Links With Parameter And Scans For Error Based SQLi)
[10] Bloggers View (Information That Bloggers Might Be Interested In)
[11] WordPress Scan (Only If The Target Site Runs On WP)
[12] Crawler
[13] MX Lookup
[A] Scan For Everything - (The Old Lame Scanner)
[F] Fix (Checks For Required Modules and Installs Missing Ones)
[U] Check For Updates
[B] Scan Another Website (Back To Site Selection)
[Q] Quit

[0] Choose Any Scan OR Action From The Above List: 13

[*] Scanning Begins ...
[1] Scanning Site: http://testphp.vulnweb.com
[5] Scan Type : MX Lookup

IP :
HOSTNAME:
```

RESULT:

The tool successfully retrieved the mail server details associated with the target domain.

USE CASE:

Used to analyze email configuration and identify mail servers during reconnaissance.

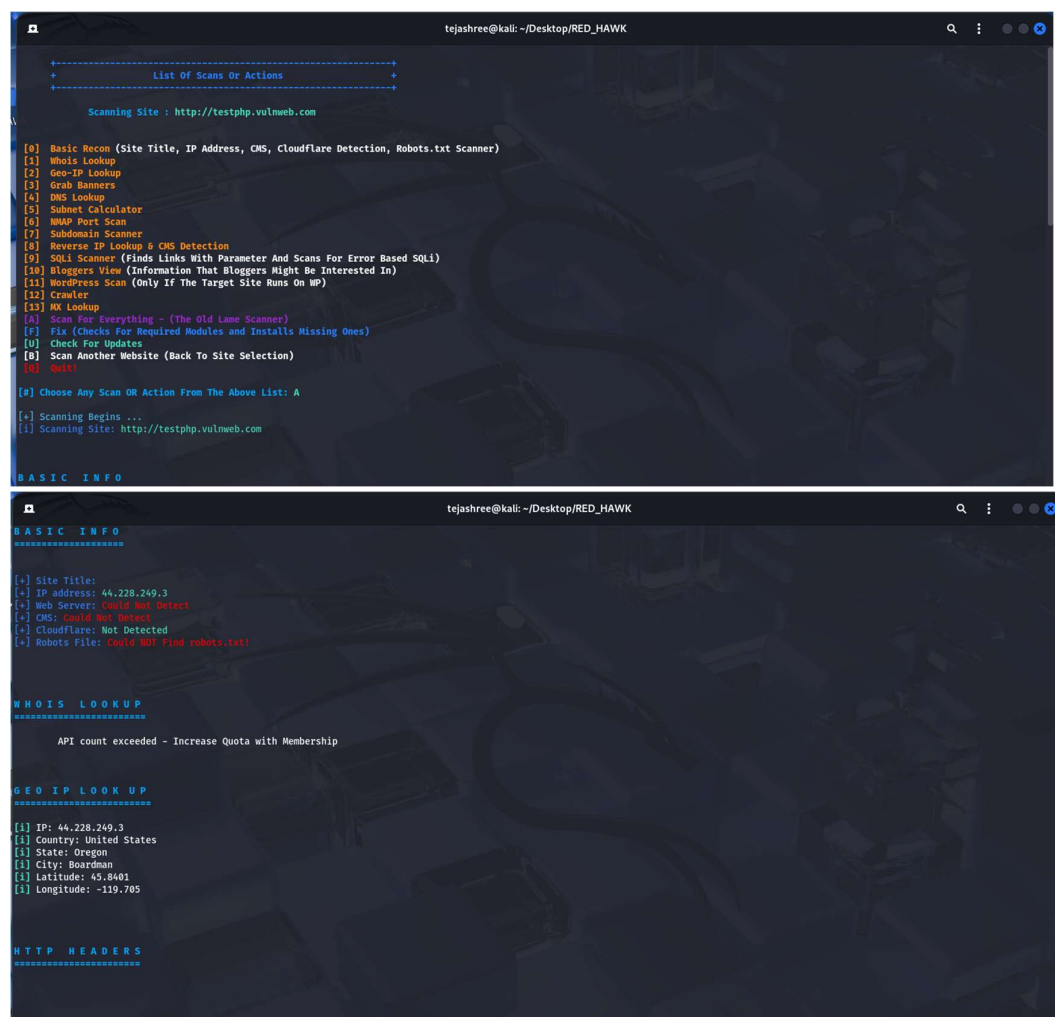
LIMITATION:

Some domains may restrict access to MX records or may not expose complete information.

15. Scan for everything

DESCRIPTION:

Scan for Everything is used to execute all available scanning features of the tool in a single run on the target website.



```
tejashee@kali: ~/Desktop/RED_HAWK

+-----+
| List Of Scans Or Actions |
+-----+

Scanning Site : http://testphp.vulnweb.com

[0] Basic Recon (Site Title, IP Address, CMS, CloudFlare Detection, Robots.txt Scanner)
[1] Whois Lookup
[2] Geo-IP Lookup
[3] Grab Banners
[4] DNS Lookup
[5] Subnet Calculator
[6] NMAP Port Scan
[7] Subdomain Scanner
[8] Reverse IP Lookup & CMS Detection
[9] SQLi Scanner (Finds Links With Parameter And Scans For Error Based SQLi)
[10] Bloggers View (Information That Bloggers Might Be Interested In)
[11] WordPress Scan (Only If The Target Site Runs On WP)
[12] Crawler
[13] MX Lookup
[14] Scan For Everything - (The Old Lane Scanner)
[P] Fix (Checks For Required Modules and Installs Missing Ones)
[U] Check For Updates
[B] Scan Another Website (Back To Site Selection)
[Q] Quit

[0] Choose Any Scan OR Action From The Above List: A

[+] Scanning Begins ...
[+] Scanning Site: http://testphp.vulnweb.com

BASIC INFO
=====
[+] Site Title:
[+] IP address: 44.228.249.3
[+] Web Server: Could Not Detect
[+] CMS: Could Not Detect
[+] CloudFlare: Not Detected
[+] Robots File: Could NOT find robots.txt!

WHOIS LOOKUP
=====
API count exceeded - Increase Quota with Membership

GEO IP LOOKUP
=====
[+] IP: 44.228.249.3
[+] Country: United States
[+] State: Oregon
[+] City: Boardman
[+] Latitude: 45.8401
[+] Longitude: -119.705

HTTP HEADERS
=====
```

```
tejasree@kali: ~/Desktop/RED_HAWK

DNS LOOKUP
=====
A : 44.228.249.3
TXT : google-site-verification:toEctYsuNIxgraKk7H3Z58PCyz2IOcc36pIupEPmVQ

SUBNET CALCULATION
=====
Address      = 44.228.249.3
Network     = 44.228.249.3/32
Netmask     = 255.255.255.255
Broadcast   = 44.228.249.3
Wildcard Mask = 0.0.0.0
Hosts Bits  = 0
Max. Hosts  = 1
Host Range  = 44.228.249.4 - 44.228.249.2
IP Class    = A
CIDR Notation = 44.228.249.3/32
IP Version  = IPv4
Is Private  = NO
DNS Hostname = ec2-44-228-249-3.us-west-2.compute.amazonaws.com

NMAP PORT SCAN
=====
error valid key required

SUB-DOMAIN FINDER
=====
```

RESULT:

The tool executed multiple scans on the target website and displayed combined results from various features. Some scans produced results, while others did not due to limitations or missing configurations.

USE CASE:

Used to perform a comprehensive reconnaissance of the target without manually selecting each scan.

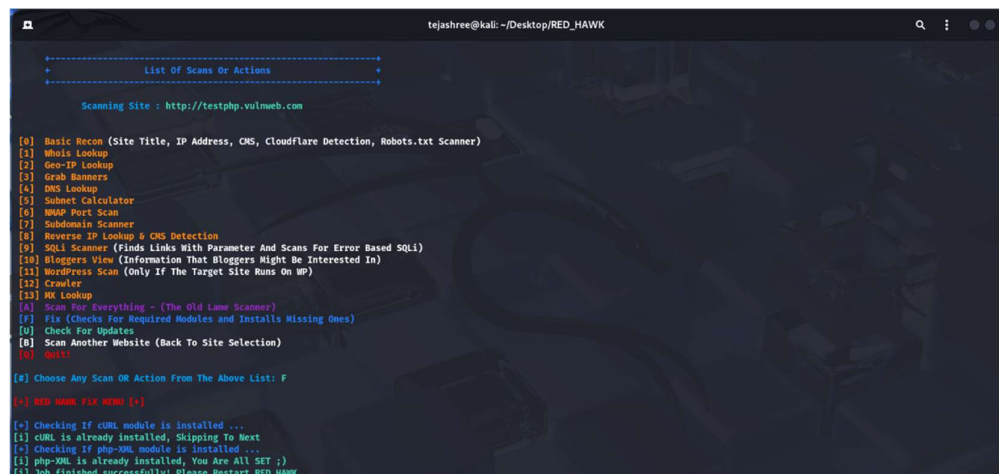
LIMITATION:

This feature may take more time to complete and some scans may not work properly due to API requirements or tool limitations.

16. Fix

DESCRIPTION:

The Fix option is used to install or repair missing dependencies required for the tool.



```
tejashee@kali: ~/Desktop/RED_HAWK
+-----+
| List Of Scans Or Actions |
+-----+

Scanning Site : http://testphp.vulnweb.com

[0] Basic Recon (Site Title, IP Address, CMS, CloudFlare Detection, Robots.txt Scanner)
[1] Whois Lookup
[2] Geo-IP Lookup
[3] Grab Banners
[4] DNS Lookup
[5] Subnet Calculator
[6] NMAP Port Scan
[7] Subdomain Scanner
[8] Reverse IP Lookup & CMS Detection
[9] SQLi Scanner (Finds Links With Parameter And Scans For Error Based SQLi)
[10] Bloggers View (Information That Bloggers Might Be Interested In)
[11] WordPress Scan (Only If The Target Site Runs On WP)
[12] Crawler
[13] HK Lookup
[A] Scan For Everything - (The Old Lane Scanner)
[F] Fix (Checks For Required Modules and Installs Missing Ones)
[U] Check For Updates
[B] Scan Another Website (Back To Site Selection)
[Q] Quit

[a] Choose Any Scan OR Action From The Above List: F

[-] RED HAWK Fix Menu [-]

[-] Checking If curl module is installed ...
[i] curl is already installed, skipping To Next
[-] Checking If php-xml module is installed ...
[i] php-xml is already installed, You Are All SET ;)
[i] Job finished successfully! Please Restart RED HAWK
```

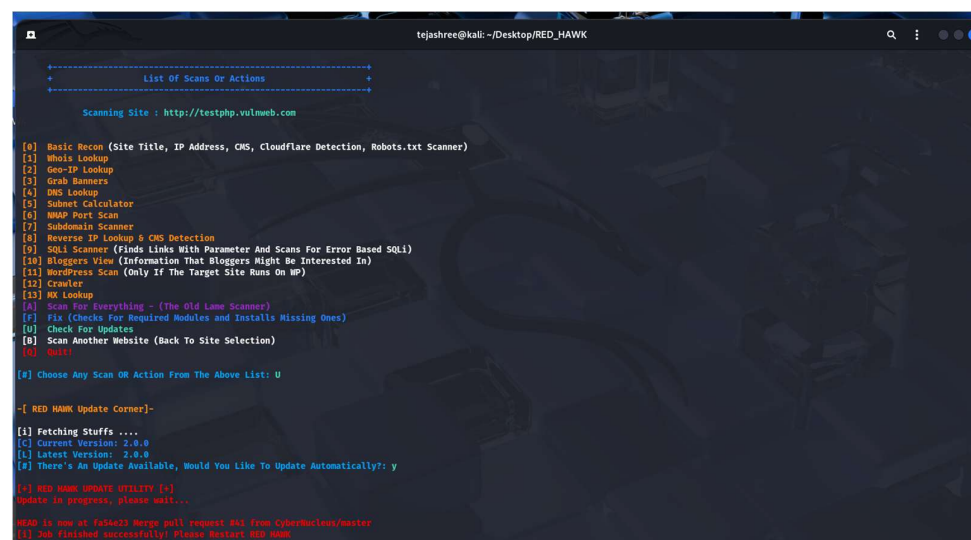
RESULT:

The tool checked for issues and attempted to resolve them.

17. Check for Updates

DESCRIPTION:

Used to check if a newer version of the tool is available.



```
tejashee@kali: ~/Desktop/RED_HAWK
+-----+
| List Of Scans Or Actions |
+-----+

Scanning Site : http://testphp.vulnweb.com

[0] Basic Recon (Site Title, IP Address, CMS, CloudFlare Detection, Robots.txt Scanner)
[1] Whois Lookup
[2] Geo-IP Lookup
[3] Grab Banners
[4] DNS Lookup
[5] Subnet Calculator
[6] NMAP Port Scan
[7] Subdomain Scanner
[8] Reverse IP Lookup & CMS Detection
[9] SQLi Scanner (Finds Links With Parameter And Scans For Error Based SQLi)
[10] Bloggers View (Information That Bloggers Might Be Interested In)
[11] WordPress Scan (Only If The Target Site Runs On WP)
[12] Crawler
[13] HK Lookup
[A] Scan For Everything - (The Old Lane Scanner)
[F] Fix (Checks For Required Modules and Installs Missing Ones)
[U] Check For Updates
[B] Scan Another Website (Back To Site Selection)
[Q] Quit

[a] Choose Any Scan OR Action From The Above List: U

[- RED HAWK Update Corner -]

[i] Fetching Stuffs ....
[C] Current Version: 2.0.0
[i] Latest Version: 2.0.0
[a] There's An Update Available, Would You Like To Update Automatically?: y

[-] RED HAWK UPDATE UTILITY [-]
Update in progress, please wait...

REDH is now at fa2a623 Merge pull request #61 from CyberNucleus/master
[i] Job finished successfully! Please Restart RED HAWK
```

RESULT:

The tool retrieved DNS records for the target website. Some information may be limited due to server restrictions.

CONCLUSION:

This experiment demonstrated the use of the Red Hawk tool for reconnaissance and basic vulnerability scanning. Various features such as DNS Lookup, Subdomain Scanning, and others were used to gather information about the target website.

Some features did not produce results due to tool limitations and API requirements. Overall, the tool provided useful insights into the structure and configuration of the target system.